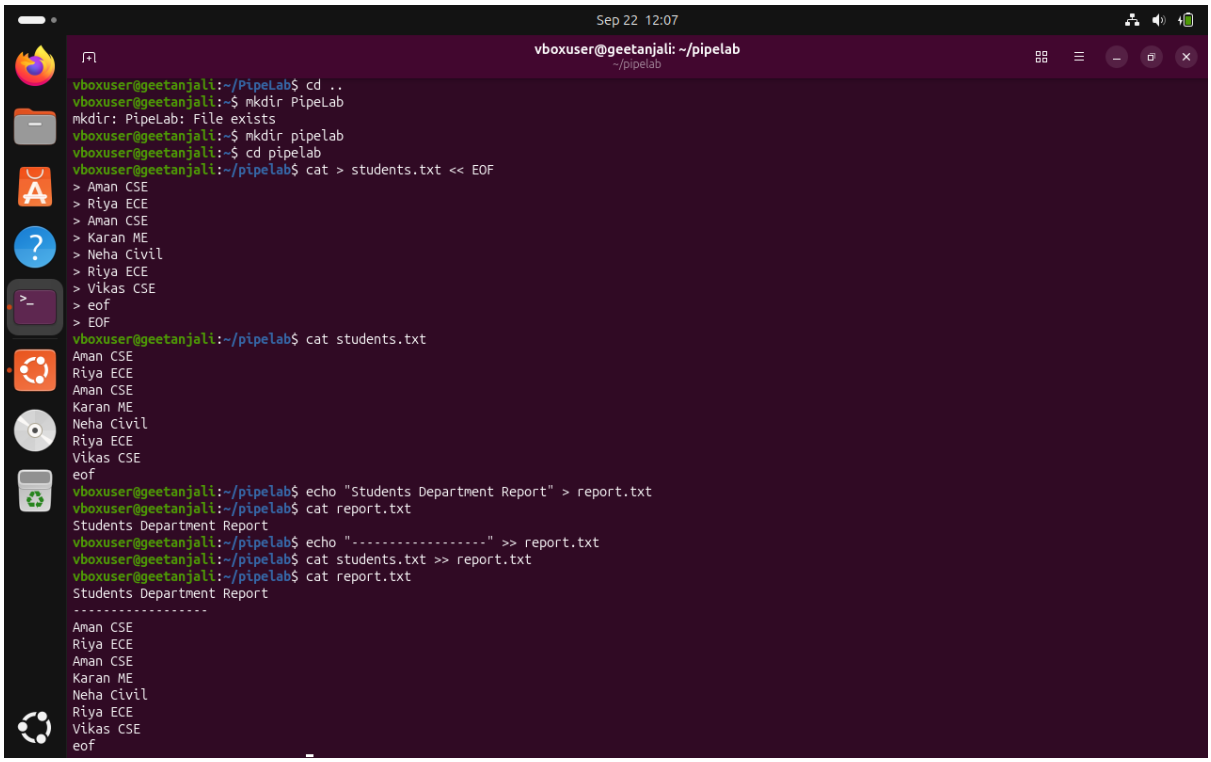
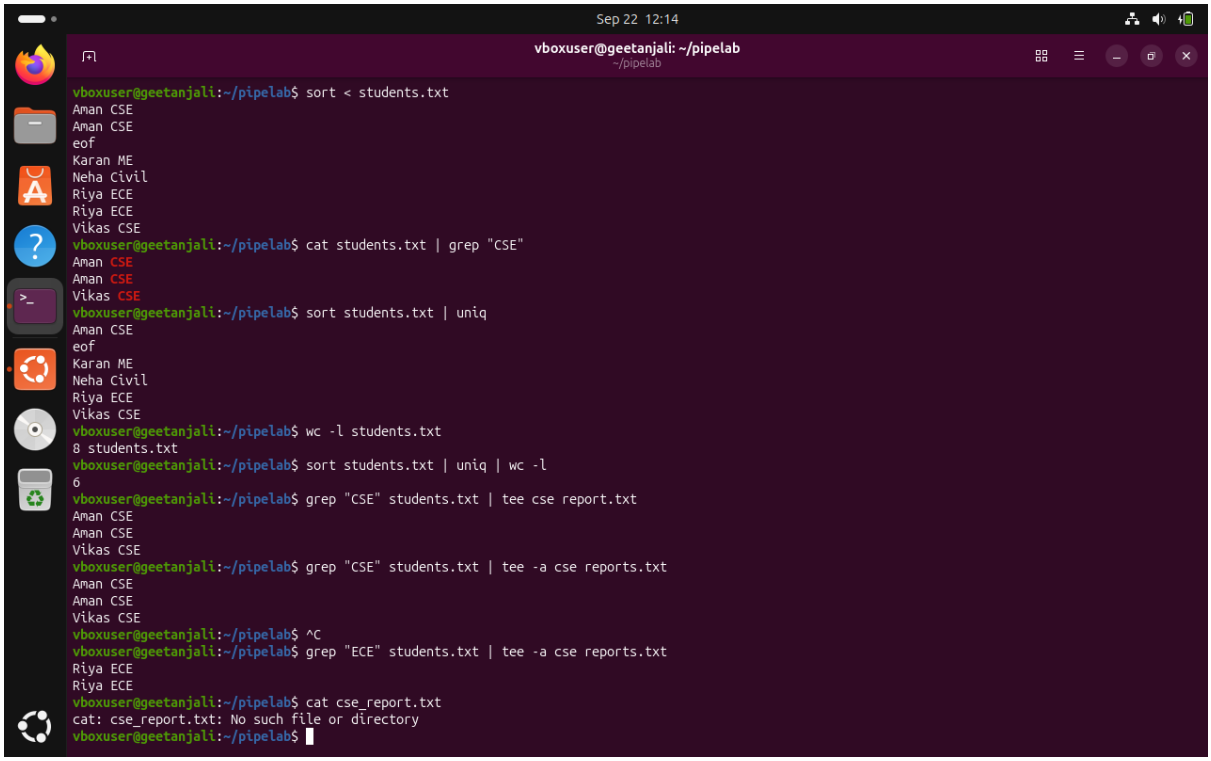
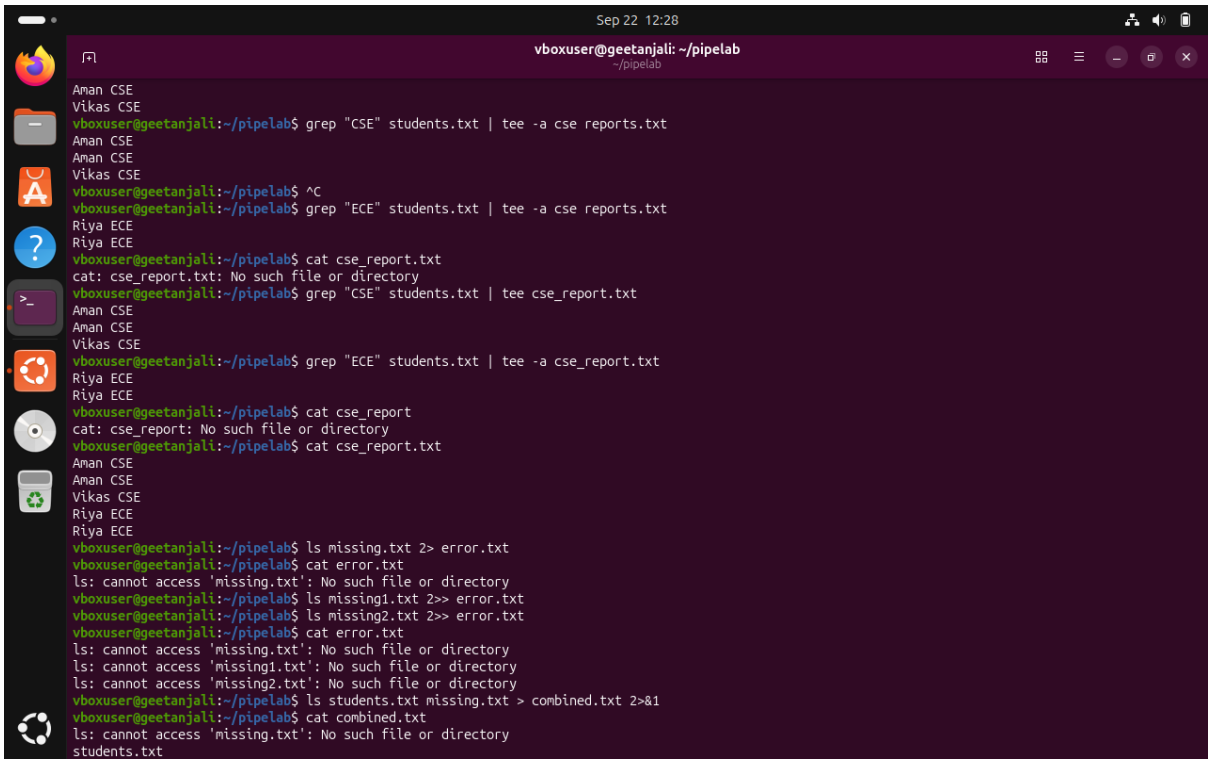
LAB ASSESSMENT 1

QUESTION 1

Create a LabFile directory and perform create/copy/move/rename/delete operations using touch, cp, mv, rm, unlinl, ls, stat.



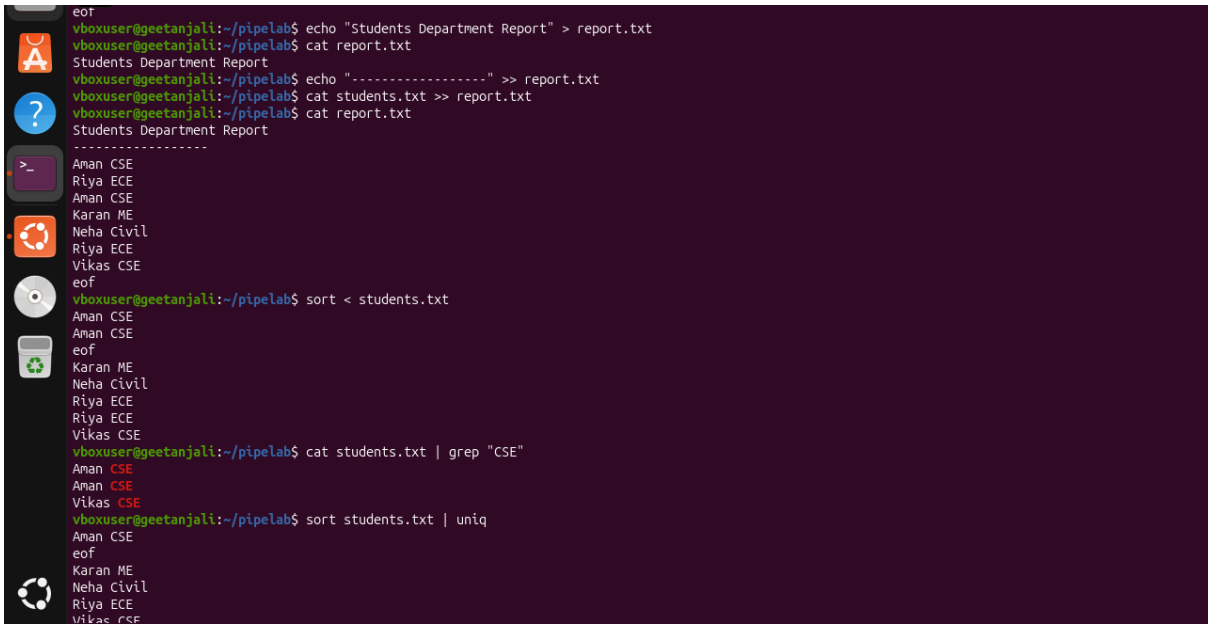

Sep 22 12:14
vboxuser@geetanjali: ~/pipelab
~/pipelab
vboxuser@geetanjali:~/pipelab$ sort < students.txt
Aman CSE
Aman CSE
eof
Karan ME
Neha Civil
Riya ECE
Riya ECE
Vikas CSE
vboxuser@geetanjali:~/pipelab$ cat students.txt | grep "CSE"
Aman CSE
Aman CSE
Vikas CSE
vboxuser@geetanjali:~/pipelab$ sort students.txt | uniq
Aman CSE
eof
Karan ME
Neha Civil
Riya ECE
Vikas CSE
vboxuser@geetanjali:~/pipelab$ wc -l students.txt
8 students.txt
vboxuser@geetanjali:~/pipelab$ sort students.txt | uniq | wc -l
6
vboxuser@geetanjali:~/pipelab$ grep "CSE" students.txt | tee cse report.txt
Aman CSE
Aman CSE
Vikas CSE
vboxuser@geetanjali:~/pipelab$ grep "CSE" students.txt | tee -a cse reports.txt
Aman CSE
Aman CSE
Vikas CSE
vboxuser@geetanjali:~/pipelab$ ^C
vboxuser@geetanjali:~/pipelab$ grep "ECE" students.txt | tee -a cse reports.txt
Riya ECE
Riya ECE
vboxuser@geetanjali:~/pipelab$ cat cse_report.txt
cat: cse_report.txt: No such file or directory
vboxuser@geetanjali:~/pipelab$

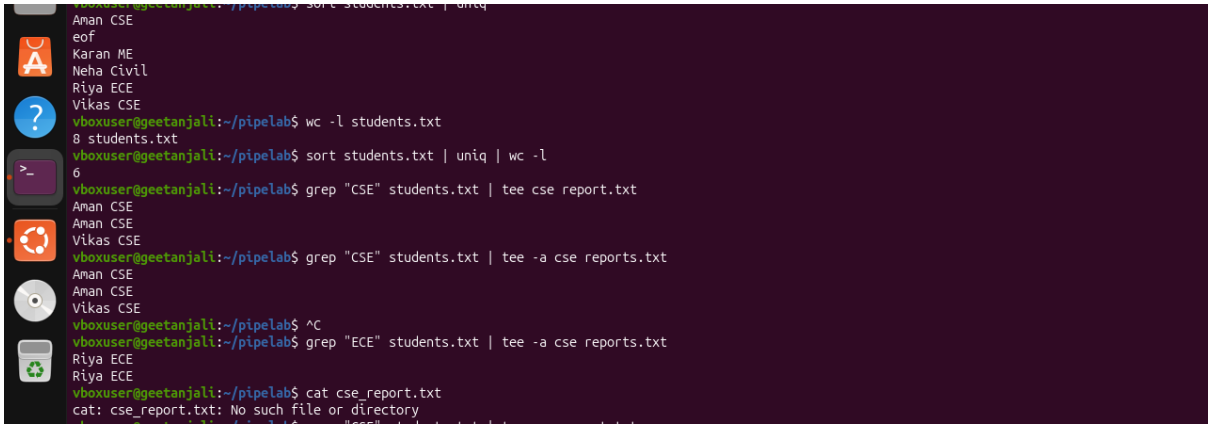

Sep 22 12:28
vboxuser@geetanjali: ~/pipelab
~/pipelab
Aman CSE
Vikas CSE
vboxuser@geetanjali:~/pipelab$ grep "CSE" students.txt | tee -a cse reports.txt
Aman CSE
Aman CSE
Vikas CSE
vboxuser@geetanjali:~/pipelab$ ^C
vboxuser@geetanjali:~/pipelab$ grep "ECE" students.txt | tee -a cse reports.txt
Riya ECE
Riya ECE
vboxuser@geetanjali:~/pipelab$ cat cse_report.txt
cat: cse_report.txt: No such file or directory
vboxuser@geetanjali:~/pipelab$ grep "CSE" students.txt | tee cse_report.txt
Aman CSE
Aman CSE
Vikas CSE
vboxuser@geetanjali:~/pipelab$ grep "ECE" students.txt | tee -a cse_report.txt
Riya ECE
Riya ECE
vboxuser@geetanjali:~/pipelab$ cat cse_report
cat: cse_report: No such file or directory
vboxuser@geetanjali:~/pipelab$ cat cse_report.txt
Aman CSE
Aman CSE
Vikas CSE
Riya ECE
Riya ECE
vboxuser@geetanjali:~/pipelab$ ls missing.txt 2> error.txt
vboxuser@geetanjali:~/pipelab$ cat error.txt
ls: cannot access 'missing.txt': No such file or directory
vboxuser@geetanjali:~/pipelab$ ls missing1.txt 2>> error.txt
vboxuser@geetanjali:~/pipelab$ ls missing2.txt 2>> error.txt
vboxuser@geetanjali:~/pipelab$ cat error.txt
ls: cannot access 'missing.txt': No such file or directory
ls: cannot access 'missing1.txt': No such file or directory
ls: cannot access 'missing2.txt': No such file or directory
vboxuser@geetanjali:~/pipelab$ ls students.txt missing.txt > combined.txt 2>&1
vboxuser@geetanjali:~/pipelab$ cat combined.txt
ls: cannot access 'missing.txt': No such file or directory
students.txt

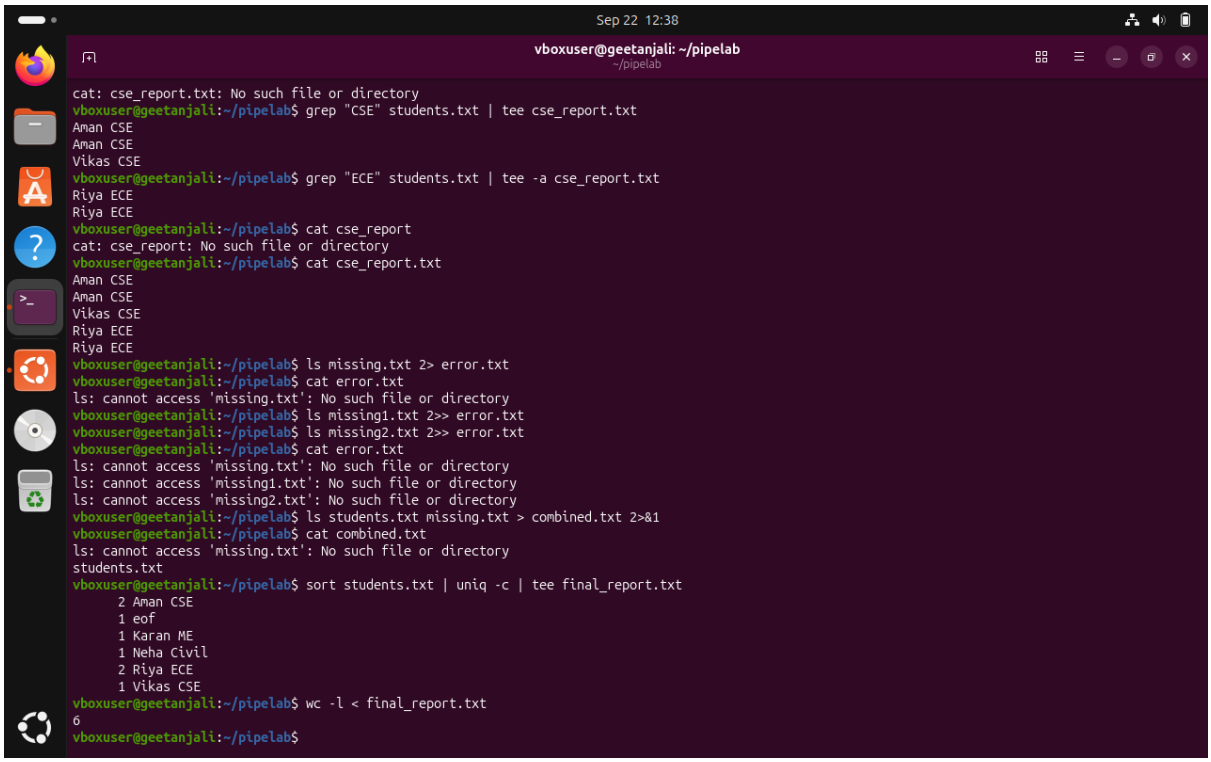
QUESTION 2

Demonstrate pipes & redirection (>,>>,<,2>,2>>,2>&1.|) with cat, grep, sort, uniq, wc, tee to generate filtered reports.



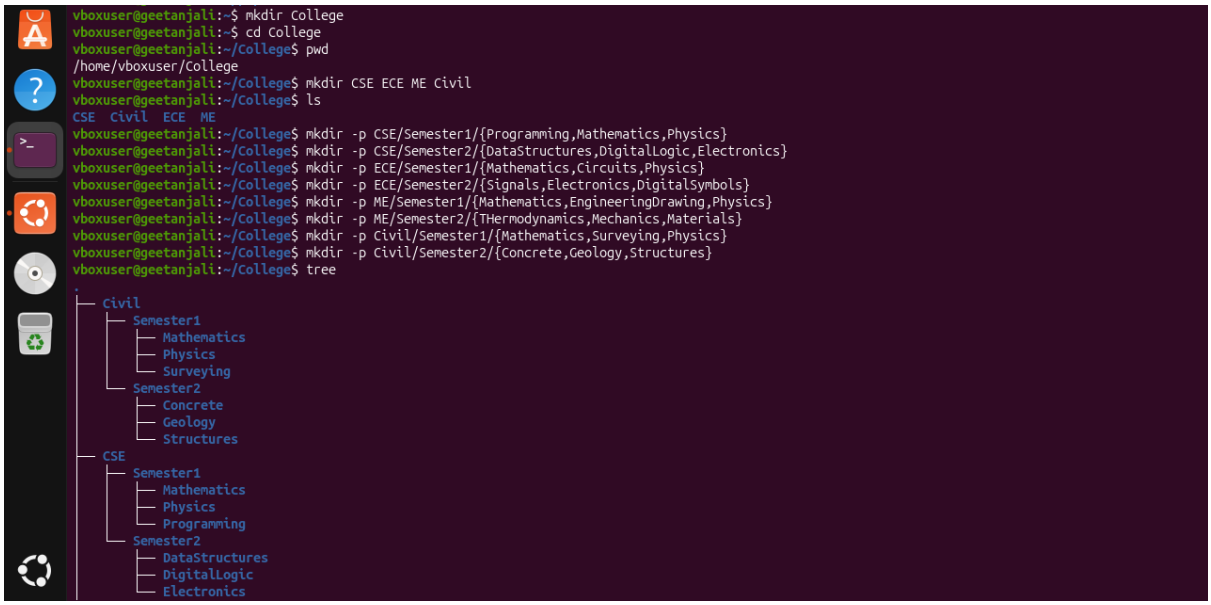


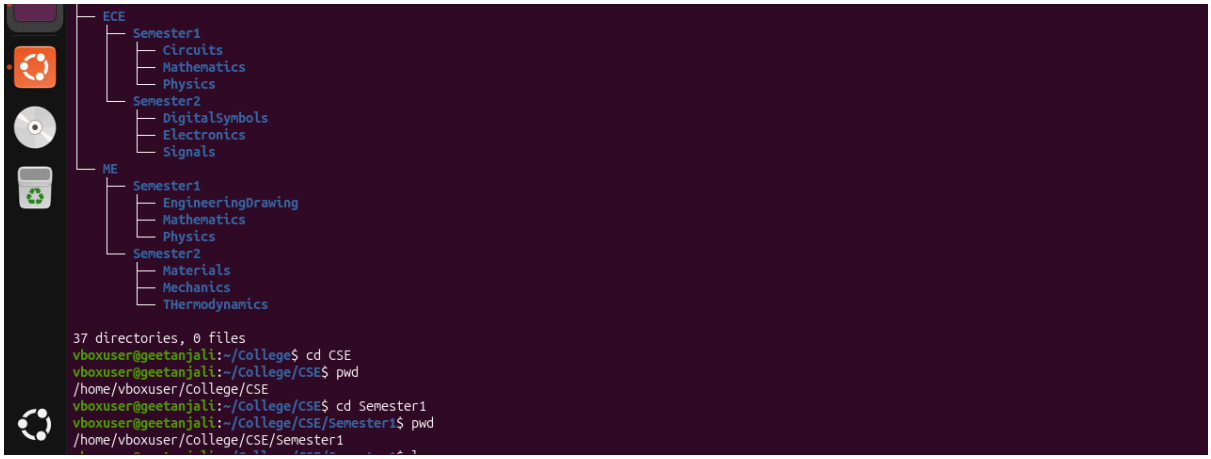

Aman CSE
eof
Karan ME
Neha Civil
Riya ECE
Vikas CSE
vboxuser@geetanjali:~/pipelab$ wc -l students.txt
8 students.txt
vboxuser@geetanjali:~/pipelab$ sort students.txt | uniq | wc -l
6
vboxuser@geetanjali:~/pipelab$ grep "CSE" students.txt | tee cse report.txt
Aman CSE
Aman CSE
Vikas CSE
vboxuser@geetanjali:~/pipelab$ grep "CSE" students.txt | tee -a cse reports.txt
Aman CSE
Aman CSE
Vikas CSE
vboxuser@geetanjali:~/pipelab$ ^C
vboxuser@geetanjali:~/pipelab$ grep "ECE" students.txt | tee -a cse reports.txt
Riya ECE
Riya ECE
vboxuser@geetanjali:~/pipelab$ cat cse_report.txt
cat: cse_report.txt: No such file or directory

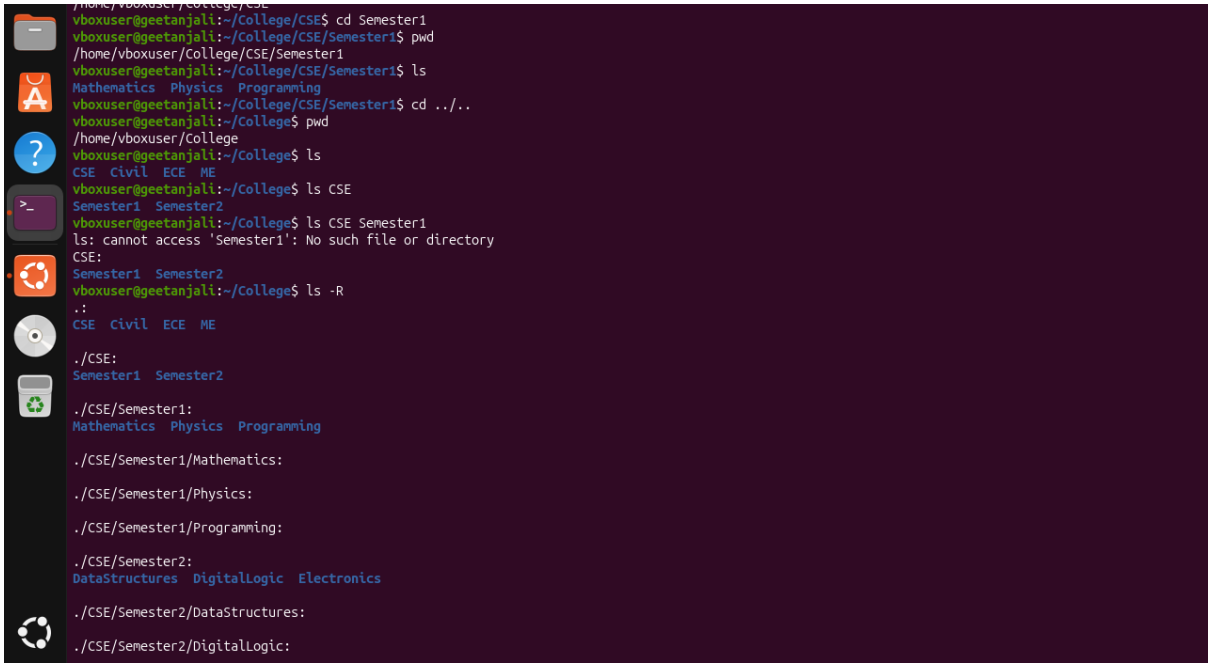

Sep 22 12:38
vboxuser@geetanjali: ~/pipelab
~/pipelab
cat: cse_report.txt: No such file or directory
vboxuser@geetanjali:~/pipelab$ grep "CSE" students.txt | tee cse_report.txt
Aman CSE
Aman CSE
Vikas CSE
vboxuser@geetanjali:~/pipelab$ grep "ECE" students.txt | tee -a cse_report.txt
Riya ECE
Riya ECE
vboxuser@geetanjali:~/pipelab$ cat cse_report
cat: cse_report: No such file or directory
vboxuser@geetanjali:~/pipelab$ cat cse_report.txt
Aman CSE
Aman CSE
Vikas CSE
Riya ECE
Riya ECE
vboxuser@geetanjali:~/pipelab$ ls missing.txt 2> error.txt
vboxuser@geetanjali:~/pipelab$ cat error.txt
ls: cannot access 'missing.txt': No such file or directory
vboxuser@geetanjali:~/pipelab$ ls missing1.txt 2>> error.txt
vboxuser@geetanjali:~/pipelab$ ls missing2.txt 2>> error.txt
vboxuser@geetanjali:~/pipelab$ cat error.txt
ls: cannot access 'missing.txt': No such file or directory
ls: cannot access 'missing1.txt': No such file or directory
ls: cannot access 'missing2.txt': No such file or directory
vboxuser@geetanjali:~/pipelab$ ls students.txt missing.txt > combined.txt 2>&1
vboxuser@geetanjali:~/pipelab$ cat combined.txt
ls: cannot access 'missing.txt': No such file or directory
students.txt
vboxuser@geetanjali:~/pipelab$ sort students.txt | uniq -c | tee final_report.txt
2 Aman CSE
1 eof
1 Karan ME
1 Neha Civil
2 Riya ECE
1 Vikas CSE
vboxuser@geetanjali:~/pipelab$ wc -l < final_report.txt
6
vboxuser@geetanjali:~/pipelab$

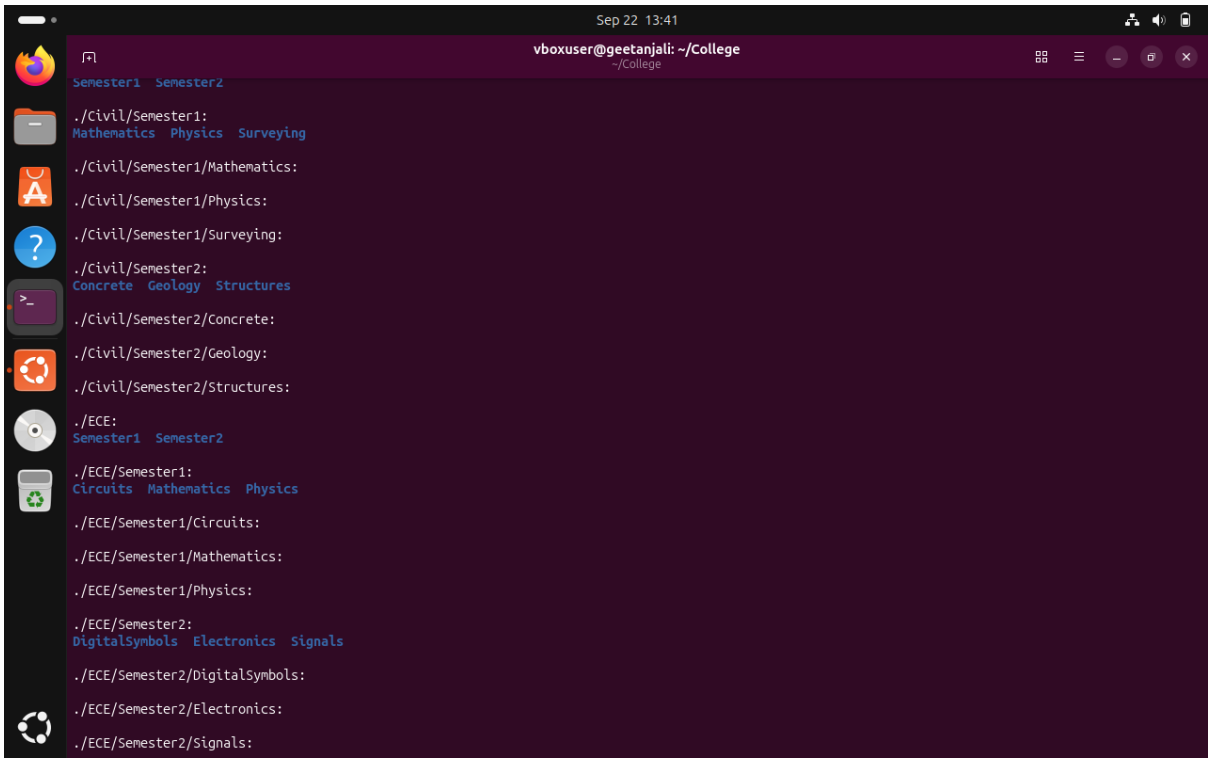
QUESTION 3

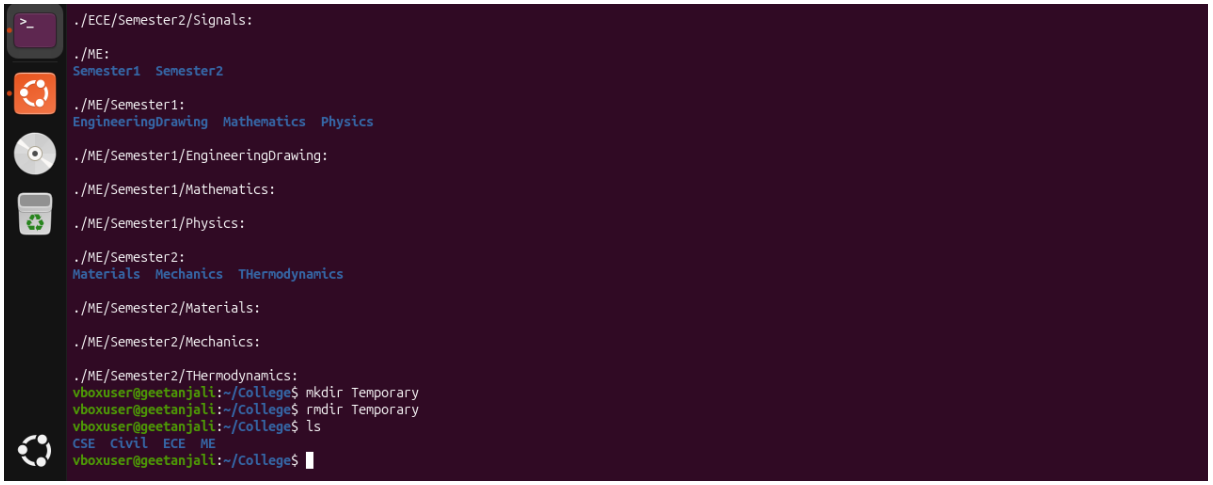
Create a College directory hierarchy with subdirectories (CSE, ECE, ME, Civil), nested semester/subject folders, and display it using tree, pwd, cd, mkdir, rmdir, ls.





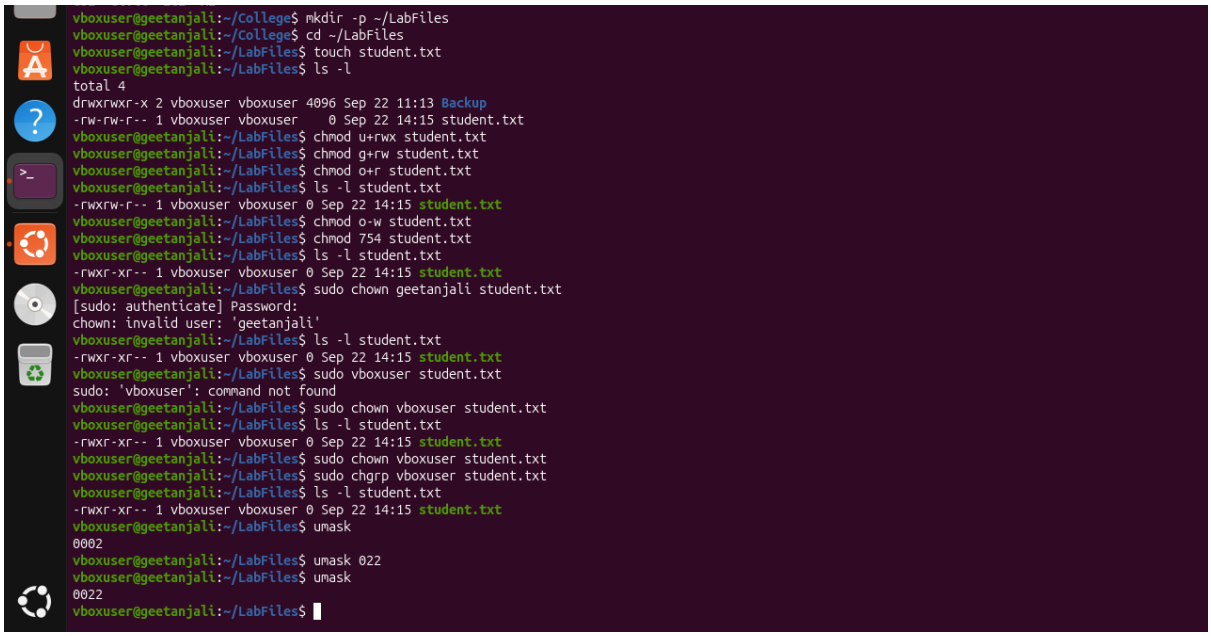

vboxuser@geetanjali:~/College/CSE$ cd Semester1
vboxuser@geetanjali:~/College/CSE/Semester1$ pwd
/home/vboxuser/College/CSE/Semester1
vboxuser@geetanjali:~/College/CSE/Semester1$ ls
Mathematics Physics Programming
vboxuser@geetanjali:~/College/CSE/Semester1$ cd ../..
vboxuser@geetanjali:~/College$ pwd
/home/vboxuser/College
vboxuser@geetanjali:~/College$ ls
CSE Civil ECE ME
vboxuser@geetanjali:~/College$ ls CSE
Semester1 Semester2
vboxuser@geetanjali:~/College$ ls CSE Semester1
ls: cannot access 'Semester1': No such file or directory
CSE:
Semester1 Semester2
vboxuser@geetanjali:~/College$ ls -R
.:
CSE Civil ECE ME
./CSE:
Semester1 Semester2
./CSE/Semester1:
Mathematics Physics Programming
./CSE/Semester1/Mathematics:
./CSE/Semester1/Physics:
./CSE/Semester1/Programming:
./CSE/Semester2:
DataStructures DigitalLogic Electronics
./CSE/Semester2/DataStructures:
./CSE/Semester2/DigitalLogic:

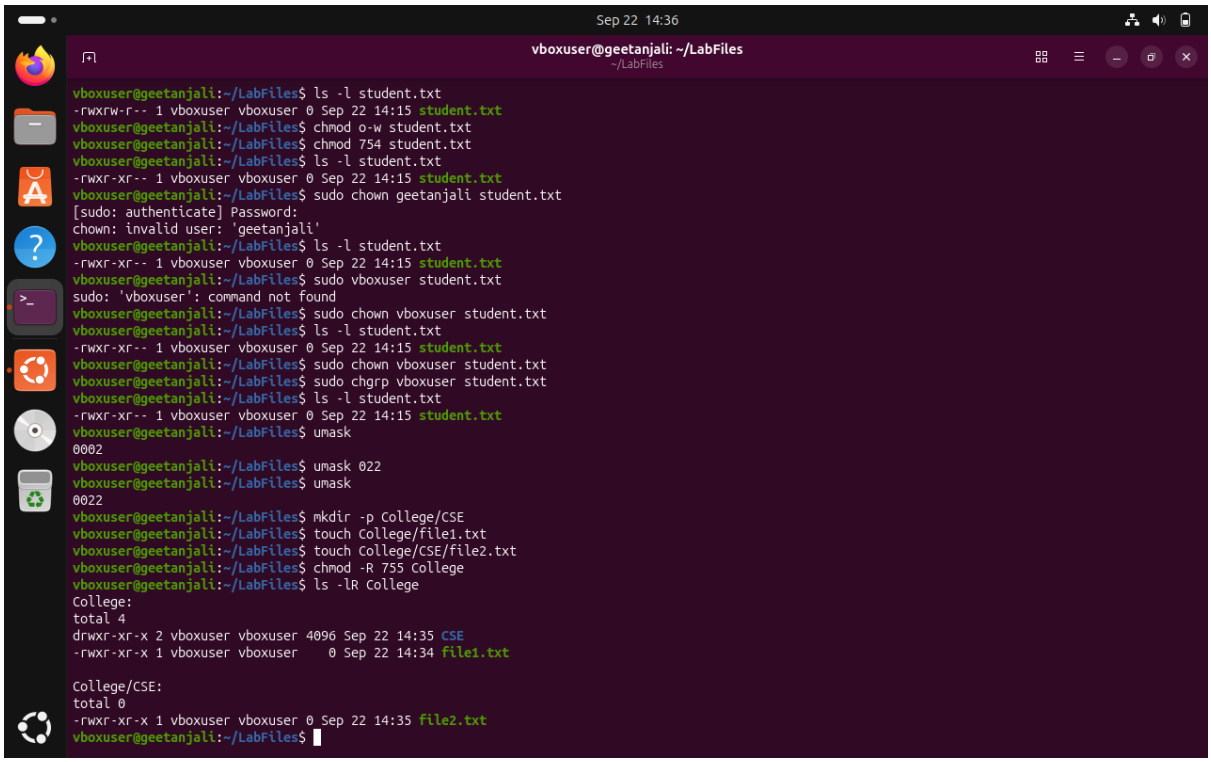

Sep 22 13:41
vboxuser@geetanjali: ~/College
~/College
Semester1 Semester2
./Civil/Semester1:
Mathematics Physics Surveying
./Civil/Semester1/Mathematics:
./Civil/Semester1/Physics:
./Civil/Semester1/Surveying:
./Civil/Semester2:
Concrete Geology Structures
./Civil/Semester2/Concrete:
./Civil/Semester2/Geology:
./Civil/Semester2/Structures:
./ECE:
Semester1 Semester2
./ECE/Semester1:
Circuits Mathematics Physics
./ECE/Semester1/Circuits:
./ECE/Semester1/Mathematics:
./ECE/Semester1/Physics:
./ECE/Semester2:
DigitalSymbols Electronics Signals
./ECE/Semester2/DigitalSymbols:
./ECE/Semester2/Electronics:
./ECE/Semester2/Signals:


./ECE/Semester2/Signals:
./ME:
Semester1 Semester2
./ME/Semester1:
EngineeringDrawing Mathematics Physics
./ME/Semester1/EngineeringDrawing:
./ME/Semester1/Mathematics:
./ME/Semester1/Physics:
./ME/Semester2:
Materials Mechanics THermodynamics
./ME/Semester2/Materials:
./ME/Semester2/Mechanics:
./ME/Semester2/THermodynamics:
vboxuser@geetanjali:~/College$ mkdir Temporary
vboxuser@geetanjali:~/College$ rmdir Temporary
vboxuser@geetanjali:~/College$ ls
CSE Civil ECE ME
vboxuser@geetanjali:~/College$

QUESTION 4

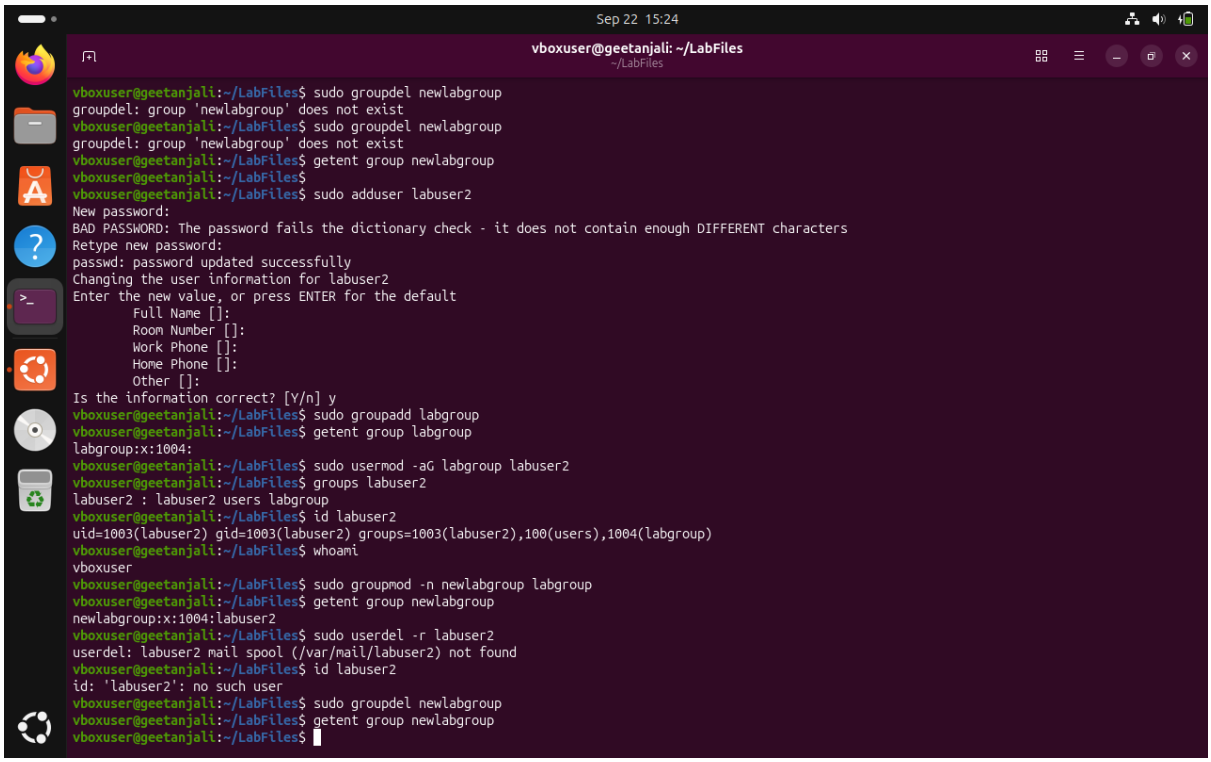
Configure file permissions & ownership (symbolic/numeric chmod, chown, chgrp, umask) and verify with ls -l including recursive updates.

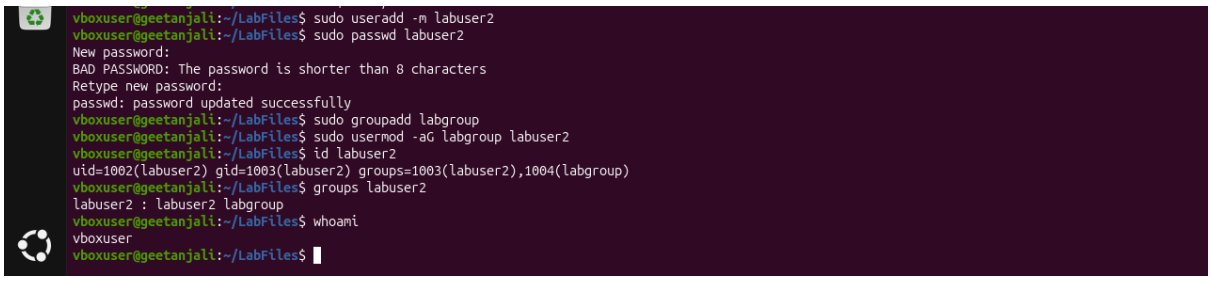




QUESTION 5

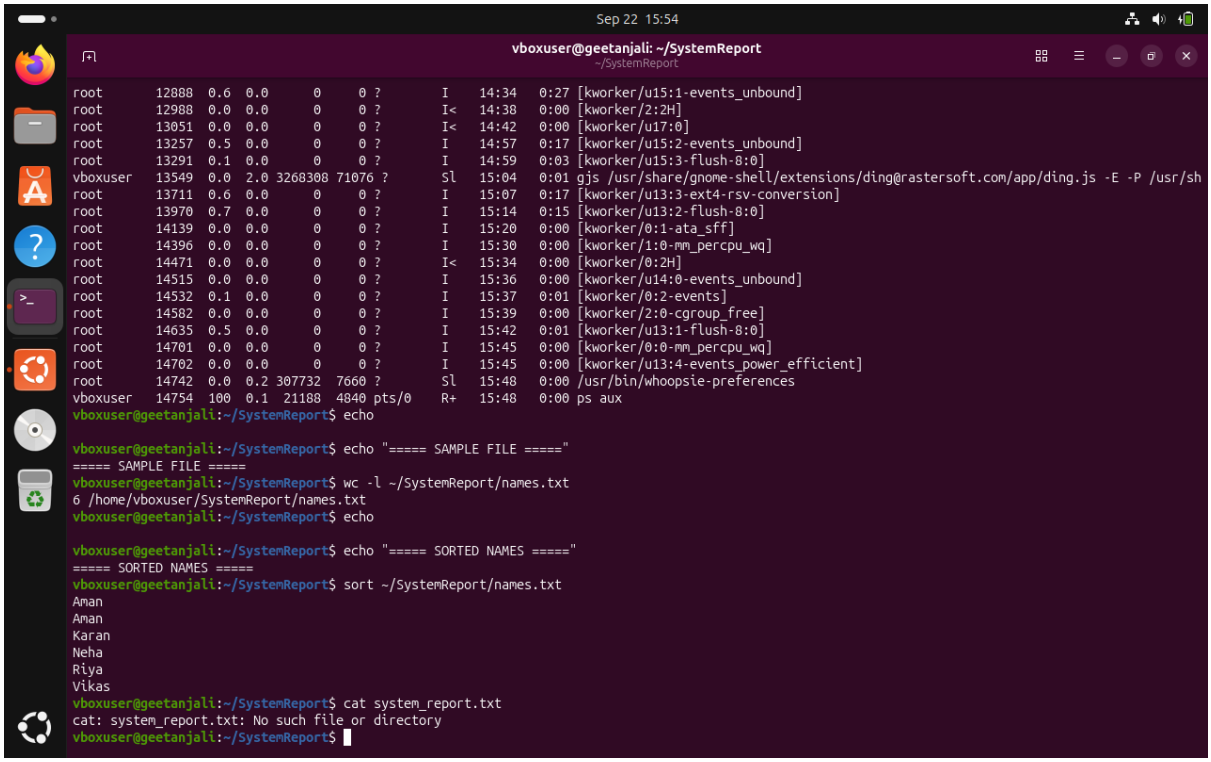
Create and manage users/groups: useradd, adduser,passwd, usermod, userdel, groupadd, groupmod, gorupdel, id, whoami.





QUESTION 6

Generate a system/directory report using find, du, df, stat, wc, sort, head, tai, uname, hostname, history, free, ops, top with screenshots and summary.



QUESTION 7

Configure File permission and ownership.

